

PMLS - Assignment 1

Start Assignment

- Due Monday by 23:59
- Points 20
- Submitting a file upload
- Available 29 Jul at 0:00 - 5 Oct at 23:59

M10 PMLS - ASSIGNMENT 1

Model Deployment Using FastAPI

Business Context:

A skin clinic conducted a marketing campaign targeting 10,000 customers. The objective is to analyze which customer segments are more likely to respond to the campaign based on demographics and purchase behavior. Insights from this analysis will help improve future targeting strategies and increase campaign effectiveness.

Tasks

1. Gender vs Campaign Response

Create a table showing the **response rate (%)** for Male and Female customers.

2. Age Group vs Campaign Response

Prepare a table showing the **response rate (%)** across the following age groups:

- <30
- 30–50
- >50

3. Purchase in Last Quarter vs Campaign Response

Construct a table showing **response rate (%)** for customers based on whether they made a purchase in the last quarter (Yes/No).

4. Product Usage vs Campaign Response

Categorize customers based on the number of unique products purchased in the last year:

- 1–4
- 5–8
- >8

Create a table showing the **response rate (%)** for each category.

Deployment Task

After completing the analysis:

1. Convert your solution into a **FastAPI application**.
2. Create an endpoint: **/campaign-analysis**
3. The endpoint should display results in a **tabular format**
4. Deploy your API using **Render**.
5. Share your **deployed API link**.

